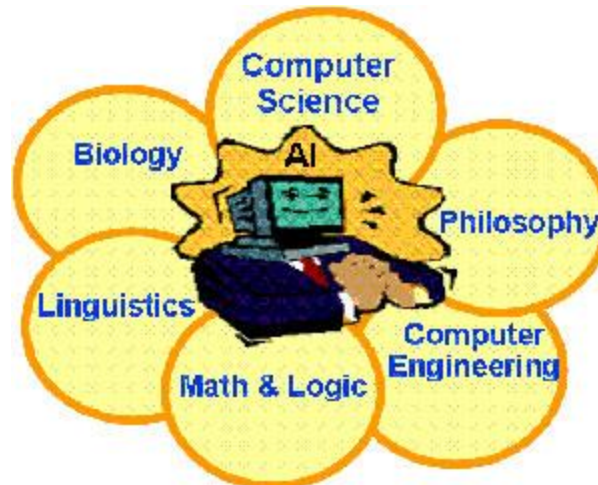


Semantics Parsing: from Language to Logic for AI

Peratham Wiriathamabhum

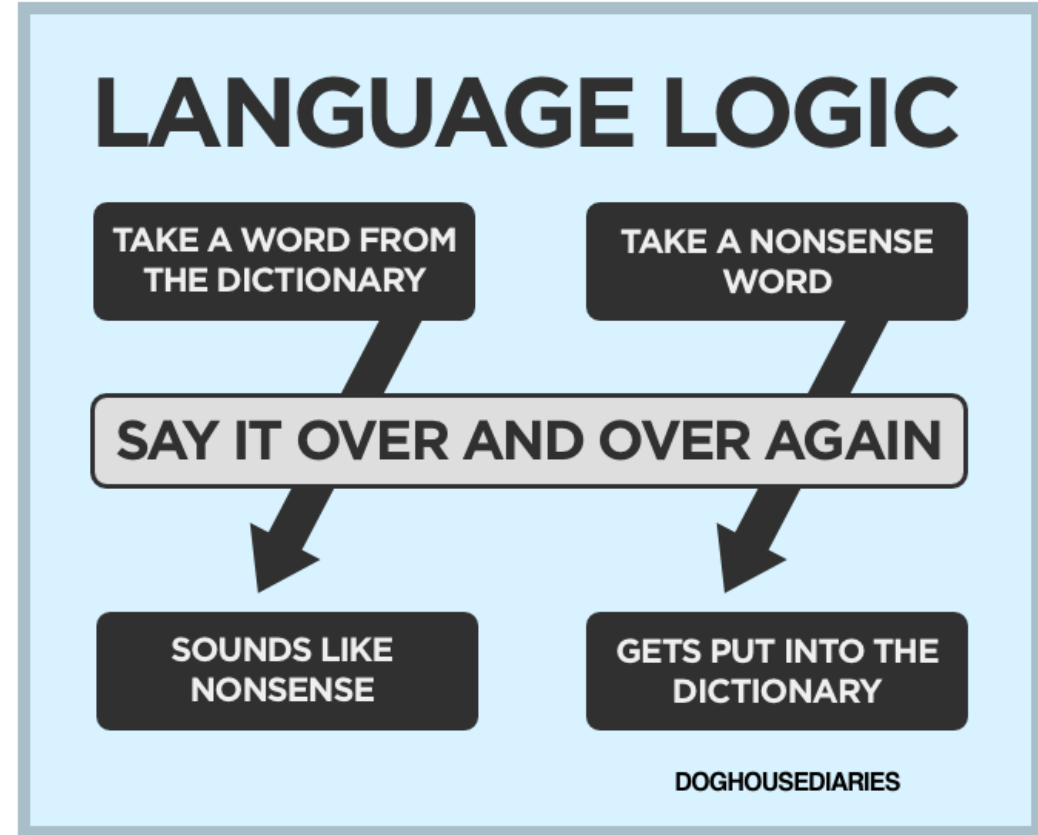
CMSC828N: Logical Aspects of AI

Fall 2014



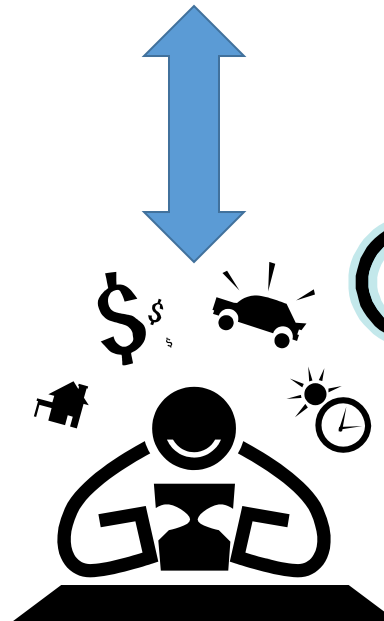
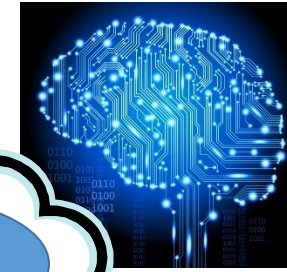
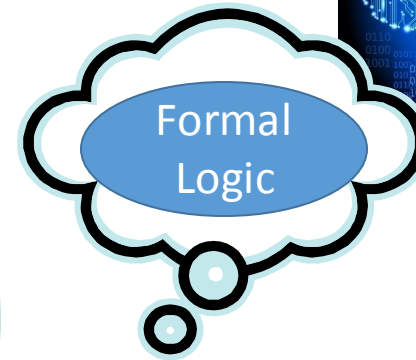
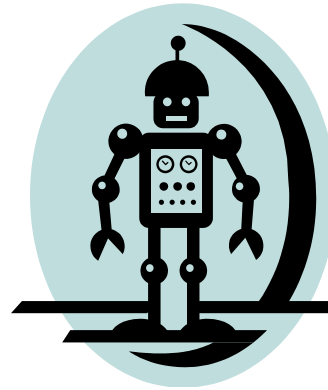
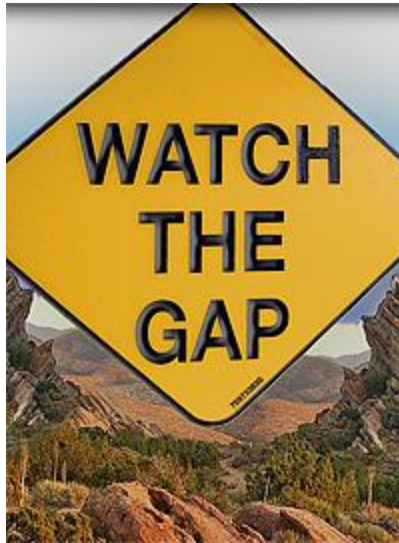
Outline

- **Language and Logic**
- Semantic Parsing
- Montague Grammar
- Recent Approaches

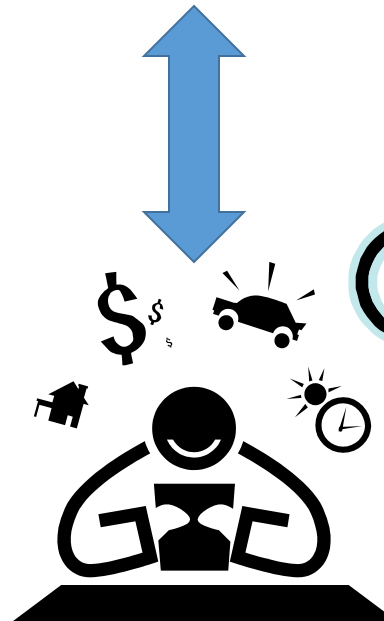
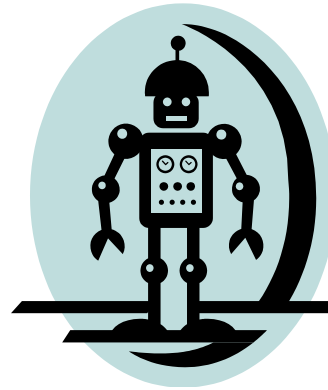


<http://thedoghousediaries.com/5347>

Language and Logic

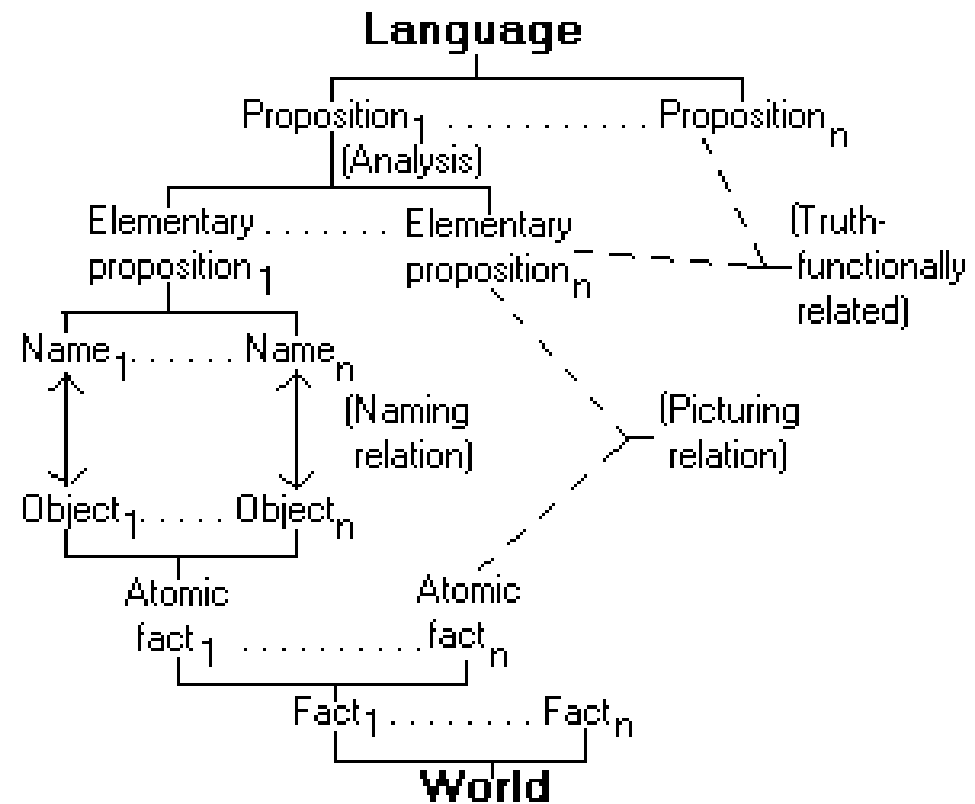


Language and Logic



Outline

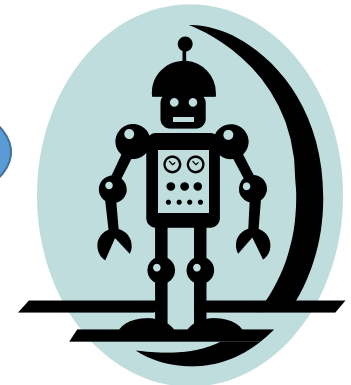
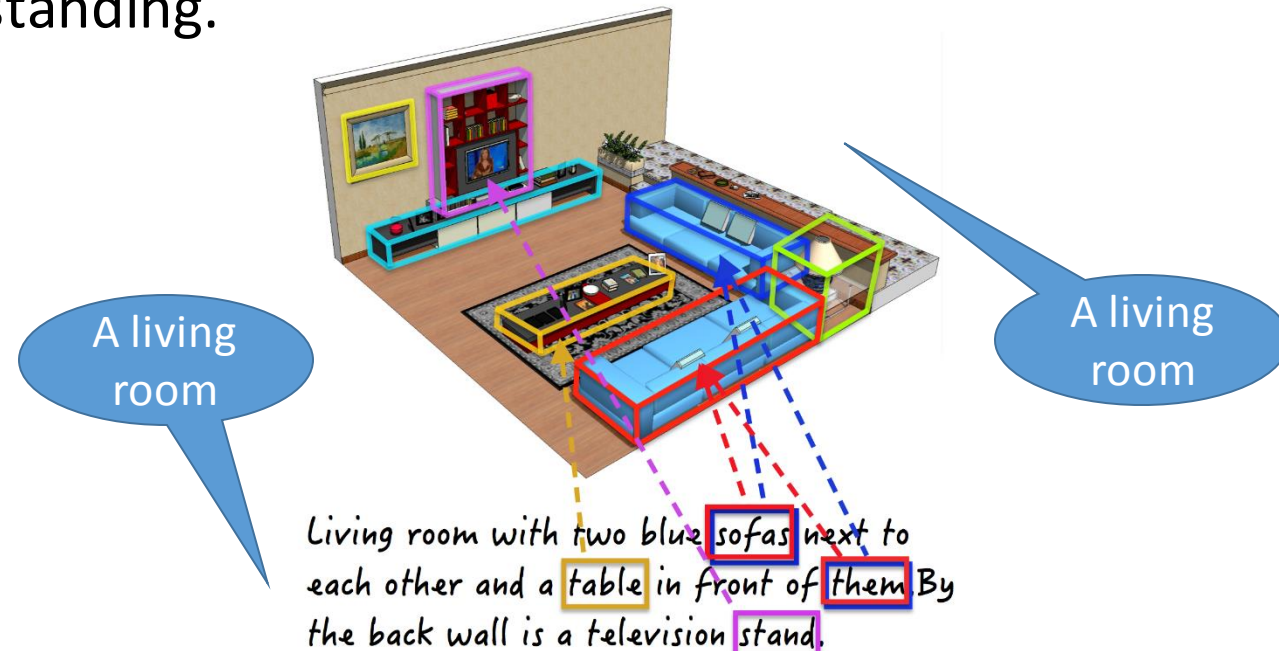
- Language and Logic
- **Semantic Parsing**
- Montague Grammar
- Recent Approaches



<http://www.jkerkkonen.com/languagelogic.htm>

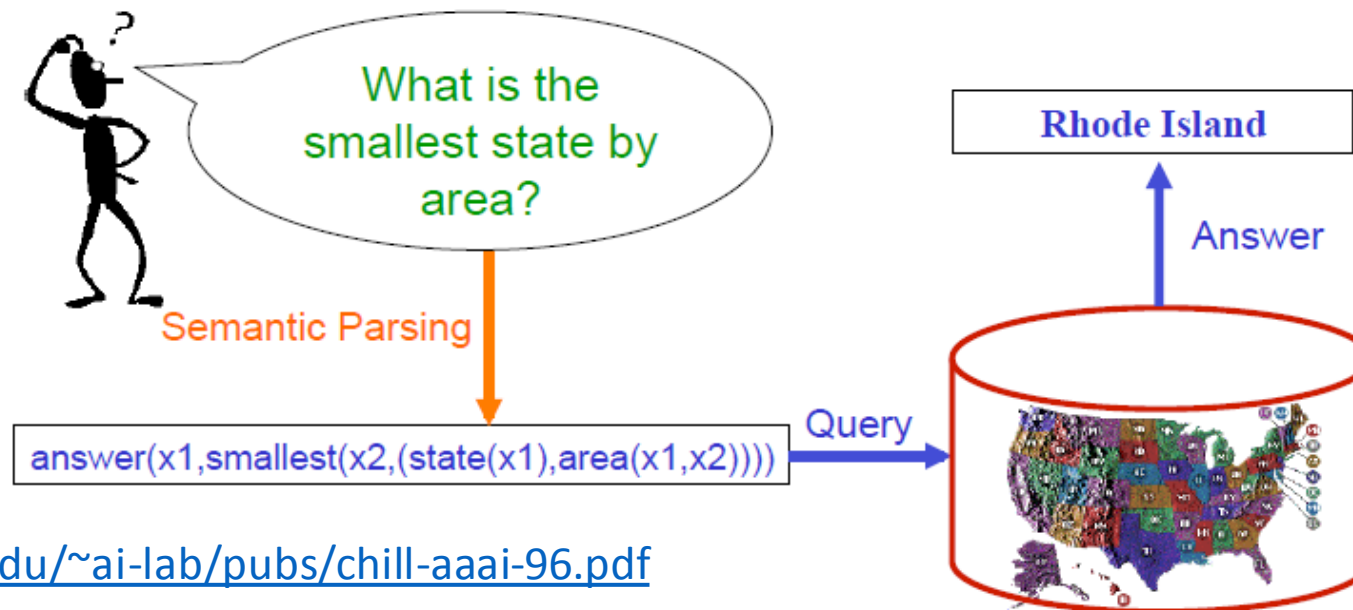
Semantic Parsing

- Toward natural language understanding.
 - Translate natural language to **some formal representations** (e.g. formal logic, ontology) that a machine can operate on (perform reasoning/execution/operation).
 - The choice of formal representation is **dependent** to the task.
 - **Sample applications:** Database Query with Natural Language Interface (next slide), Scene Understanding.



Natural Language Interface for Database Query

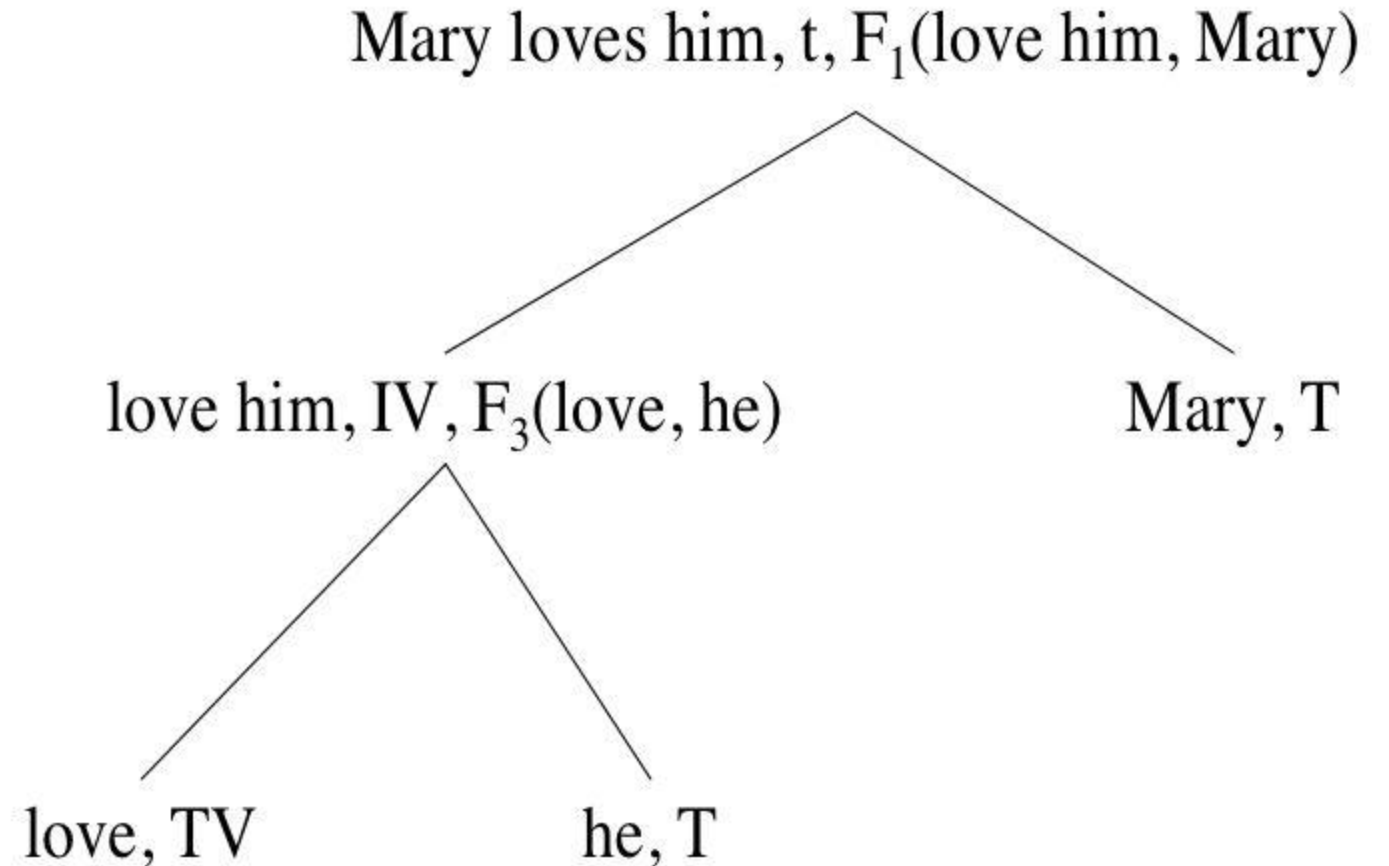
- In early 1970-80s, *semantic grammars* were done **manually**.
- [Zelle and Mooney, 1996] proposed the CHILL system which uses an inductive logic programming (ILP) to **learn** the grammar that maps natural language sentences to query parses.
 - The output is a shift-reduce parser controlled by a prolog program.



Good: Automatic
Bad: No lexicon learning
(predefined)

Outline

- Language and Logic
- Semantic Parsing
- **Montague Grammar**
- Recent Approaches



http://www-personal.umich.edu/~akao/NLP_Paper.htm

Montague Logic/Grammar/Semantics

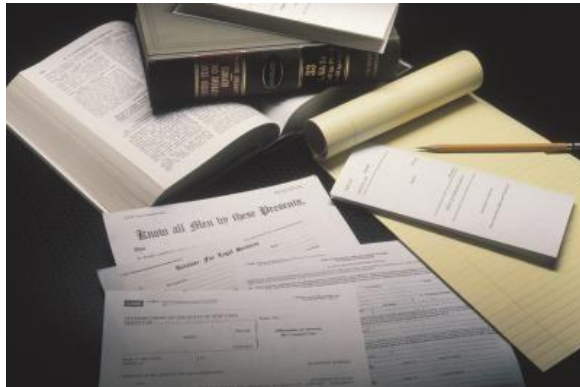


- Richard Montague (1930-1971)
 - He is a student of Alfred Tarski.
 - He taught at UCLA.
 - He proposed a method to map natural language to first-order predicate logic using lambda calculus and compositional semantics.
 - His inspiring idea:

“I reject the contention that an important theoretical difference exists between formal and natural languages.”

Montague Logic/Grammar/Semantics

- Maps syntactic structure with semantic structure using types.
- Use formal language to describe natural language.
- Was described in 3 main papers:
 - Universal Grammar (UG - 1970) – syntax and semantics
 - English as a Formal Language (1970)
 - The Proper Treatment of Quantification in Ordinary English (PTQ - 1973) - applications



λ -calculus
Generalized Quantifier
Tense and Modal Logic
<All the rest>

Montague Logic/Grammar/Semantics

- Syntax

- **“The purpose of syntax is to produce the input for the semantics.”**
- Rules: syntactic categories defined as basic or recursive clauses.
 - 2 categories: Sentence category t (output of recursive rules) and entity category e .
 - Categories are of the form t/e which is semantics of an entity e into the truth value of a sentence t .
 - More slashes for new categories, e.g. t/e , $t//e$, ...
- Operations: functions that defines how categories form new phrases.
- Are applied to a lexicon to define the syntax of a fragment of English.

Montague Logic/Grammar/Semantics

- Rules

- There are 2 types of rules which every non-primitive type contains lexical rules and recursive rules.
- Lexical rules state the category of a lexical phrase.
- Recursive rules define a function $F(x, y)$ that creates a phrase in X from a phrase in X/Y and a phrase in Y .
- Most recursive rules are just concatenation $F(x, y) = xy$.
does not work with determiners

Montague Logic/Grammar/Semantics

- Syntactic Categories

Category	Abbreviation	PTQ Name	Nearest linguistic equivalent
t	(primitive)	Truth-value expression; or declarative sentence	Sentence
e	(primitive)	Entity expression; or individual expression	(noun phrase)
t/e	IV	Intransitive verb phrase	transitive verb, transitive verb and its object, or other verb phrases
t/IV	T	Term	Noun phrase
IV/T	TV	Transitive verb phrase	Transitive verb
IV/IV	IAV	IV-modifying adverb	VP-adverb and prepositional phrases containing in and about.
t/e	CN	Common noun phrase	Noun or NOM
t/t	None	Sentence-modifying adverb	Sentence-modifying adverb
IAV/T	None	IAV-making preposition	Locative, etc., preposition
IV/t	None	Sentence-taking verb phrase	V which takes that-COMP
IV/IV	None	IV-taking verb phrase	V which takes infinitive COMP

Montague Logic/Grammar/Semantics

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IV/t	None	Sentence-taking verb phrase	V which takes that-COMP
IV/IV	None	IV-taking verb phrase	V which takes infinitive COMP

$B_{IV} = \{\text{run, walk, talk, rise, change}\}$

$B_T = \{\text{John, Mary, Bill, ninety, he}_0, \text{he}_1, \text{he}_2, \dots\}$

$B_{TV} = \{\text{find, lose, eat, love, date, be, seek, conceive}\}$

$B_{IAV} = \{\text{rapidly, slowly, voluntarily, allegedly}\}$

$B_{CN} = \{\text{man, woman, park, fish, pen, unicorn, price, temperature}\}$

$B_{t/t} = \{\text{necessarily}\}$

$B_{IAV/T} = \{\text{in, about}\}$

$B_{IV/t} = \{\text{believe that, assert that}\}$

$B_{IV//IV} = \{\text{try to, wish to}\}$

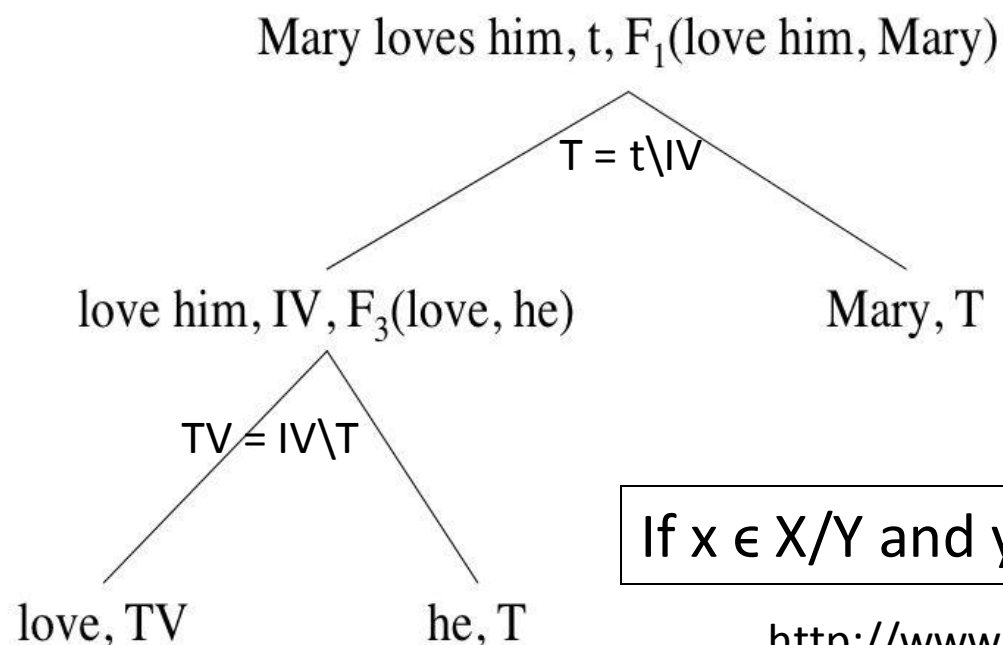
http://www-personal.umich.edu/~akao/NLP_Paper.htm

<http://www.cs.rhul.ac.uk/~zhaohui/montague73.pdf>

Montague Logic/Grammar/Semantics

- Recursive Rules

Example rule that transform 'he' to 'him'
for a transitive verb.



$F_3(x, y) =$

If first word of x is a TV (transitive verb):

$x y$ if y is not a variable he

$x \text{ him}_i$ if y is he_i

If x is $x_1 x_2$ where x_1 is a TV/T (transitive verb / term):

$x_1 y x_2$ if y is not a variable

$x \text{ him}_i x_2$ if y is he_i

$F_3(\text{shave}, \text{a fish}) = \text{shave a fish}$

$F_3(\text{seek}, \text{he}_1) = \text{seek him}_1$

$F_3(\text{read a large book}, \text{Mary}) = \text{read Mary a large book}$

If $x \in X/Y$ and $y \in Y$ then $F(x, y) \in X$

Montague Logic/Grammar/Semantics

- Semantics

- Derived from intensional logic.
- Intensions and Extensions
 - Extension: Semantic interpretation
 - Intension: **Function** generating an extension
 - Index: An input to an intension consists of variables about the context of the expression

Extension Category	Appropriate Intension Function
Truth-value of a sentence	From indices to truth-values
Thing named by a name	From indices to things
Set of objects a common noun or intransitive verb phrase applies to	From indices to sets

Montague Logic/Grammar/Semantics

- Extension

- An extension of the name '**John**' is the individual named '**John**'.
- An extension of the set of objects '**student**' is the set of all '**students**'.
- An extension of the set of objects '**read quickly**' is the set of all individual who '**read quickly**'.
- An extension of a truth-value depends on the world identified, for example, '**John read quickly**' has an extension which is based on the extension of '**John**' and the extension of '**read quickly**'.

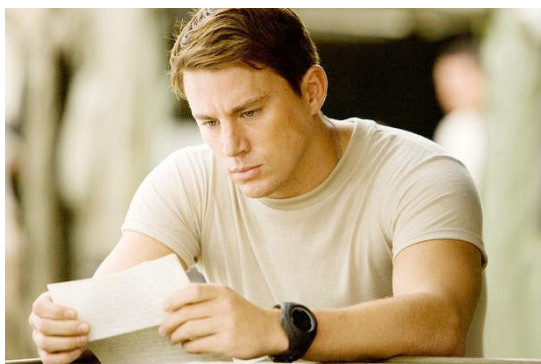
- Problem:

Classmate of John	=	Graduate student at U Mich
Former [Classmate of John]	◦	Former [Graduate student at U Mich]

Intensions matter.

http://www-personal.umich.edu/~akao/NLP_Paper.htm

http://www.aceshowbiz.com/still/00004059/dear_john11.html



Montague Logic/Grammar/Semantics

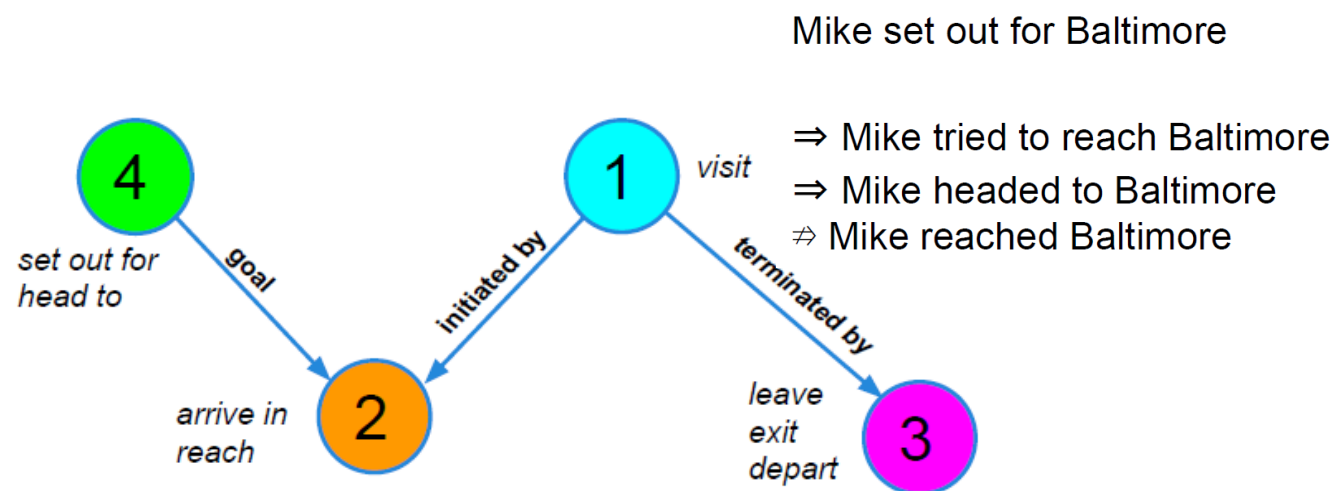
- Intensional Logic (IL) [Montague 1973; Lewis and Steedman 2014]
 - A sentence as a function from possible worlds and moments of time to truth values.

Mike set out for Baltimore

$\exists e[\text{rel4}(x,y,e) \wedge \diamond \exists e'[\text{rel2}(\text{mike}, \text{baltimore}, e') \wedge \text{goal}(e,e')]$

Mike tried to reach Baltimore

$\diamond \exists e'[\text{rel2}(\text{mike}, \text{baltimore}, e') \wedge \text{goal}(e,e')]$



- Define an intension by writing a function and index of an extension's expression.

A caret symbol $\wedge$ is prefixed to an expression to indicate that an expression's extension is its intension.

- Example: if $\wedge j$ represents the individual **John**, then $\wedge j$ is a function that finds an individual named **John** at any given world and time.

Montague Logic/Grammar/Semantics

- Translation of a syntactic rule to IL types.
 - If $x \in X/Y$, $y \in Y$ and x, y translates to $x'y'$ then $F(x, y)$ translates into $x'(\wedge y')$.
 - $\wedge y'$ refers to the intension of y' .
 - IL's types are based on t and e .
 - $\langle a, b \rangle$ means a function from type a to type b .
 - $\langle s, a \rangle$ means a function from world-time pair s to type a . (used to define an intension and extension for a lexicon.)
 - X/Y has the type $\langle \langle s, y \rangle, x \rangle$.
 - Possible denotations (Cartesian product) D , individual A , world W , time T .

#	Rule	Semantic Rule	Semantic Set
1	t	D_t is $\{0,1\}$.	all truth-values
2	e	D_e is A .	all entities
3	If a and b are types, $\langle a, b \rangle$ is a type	$D_{\langle a, b \rangle}$ is $D_b^{D_a}$	all functions from D_a to D_b for D_a
4	If a is a type, $\langle s, a \rangle$ is a type	$D_{\langle s, a \rangle}$ is $S_a = S_a^{W \times T}$	all functions from world-time pairs to D_a

Montague Logic/Grammar/Semantics

- General Quantification (GQ)
 - General Quantification allows syntax and semantics structures to be equivalent.
 - 'John talks' should be equivalent to 'Every student talks'.

Sentence	First Order Logic	General Quantification
John talks	Talks(j)	John' ($\wedge$ talk')
Every student talks	$\forall x [\text{Student}(x) \rightarrow \text{Talks}(x)]$	Every man' ($\wedge$ talk')

If α translates into α' , then

$F_0(\alpha)$ translates into $P'[(\forall x)(\alpha'(x) \rightarrow P\{x\})]$

$F_1(\alpha)$ translates into $P'[(\exists y) ((\forall x) [\alpha'(x) \leftrightarrow x = y] \wedge P\{y\})]$

$F_2(\alpha)$ translates into $P'[(\exists x)(\alpha'(x) \wedge P\{x\})]$

Equation 4: Translation rule for quantifier phrases

If $\alpha \in P_{CN}$ then $F_0(\alpha), F_1(\alpha), F_2(\alpha) \in P_T$, where

$F_0(\alpha) = \text{every } \alpha$

$F_1(\alpha) = \text{the } \alpha$

$F_2(\alpha) = \text{a/an } \alpha$

Equation 3: Syntactic Rule for quantifier Phrases

Montague Logic/Grammar/Semantics

- Model Theoretic Semantics

- Define the truth value for base cases and compose the truth value of the whole corresponding to rules used in interpreting intensions of a sentence fragment.

- Models are indices.

Base Rule:	α satisfies F
Recursive Rule:	α satisfies “F and G” iff α satisfies F and α satisfies G

Figure 6: Truth Definitions

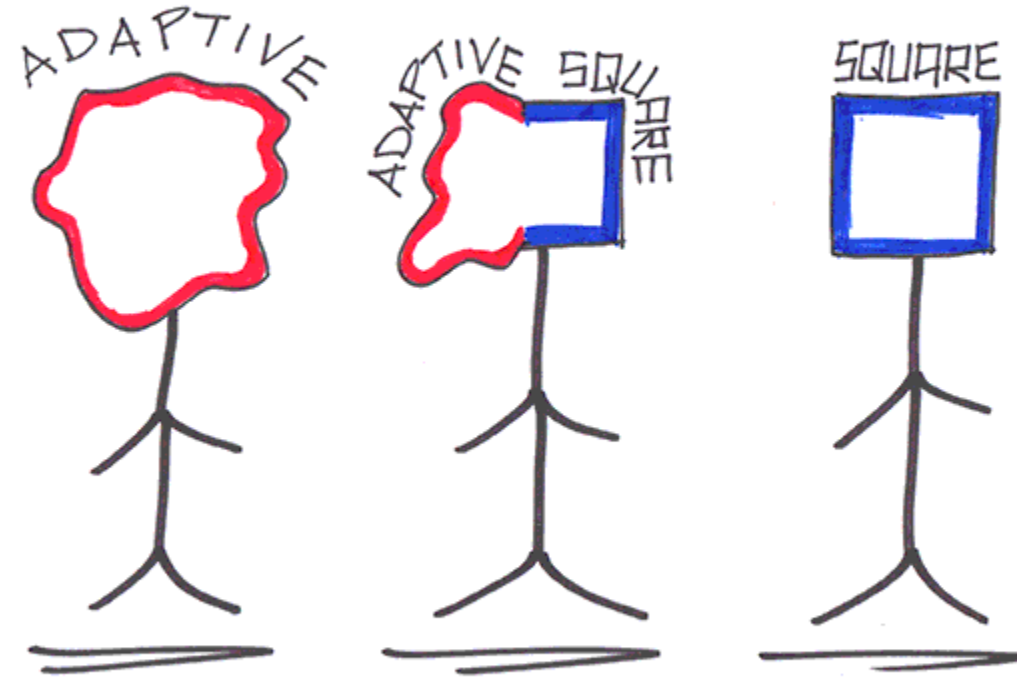
Concept	Truth Definition
Consequence	X is a consequence of sentences in class K iff X is true in every model where every sentence in K is true.
Equivalence	X and Y are logically equivalent if X is a consequence of Y and Y is a consequence of X
Truth	X is logically true if for all models, X is true
Contradiction	Class K is contradictory if no model exists where all sentences in K are true

Table 5: Truth definitions for semantic concepts

Outline

- Language and Logic
- Semantic Parsing
- Montague Grammar
- **Recent Approaches**

THINKING APPROACHES



STUART PATTON

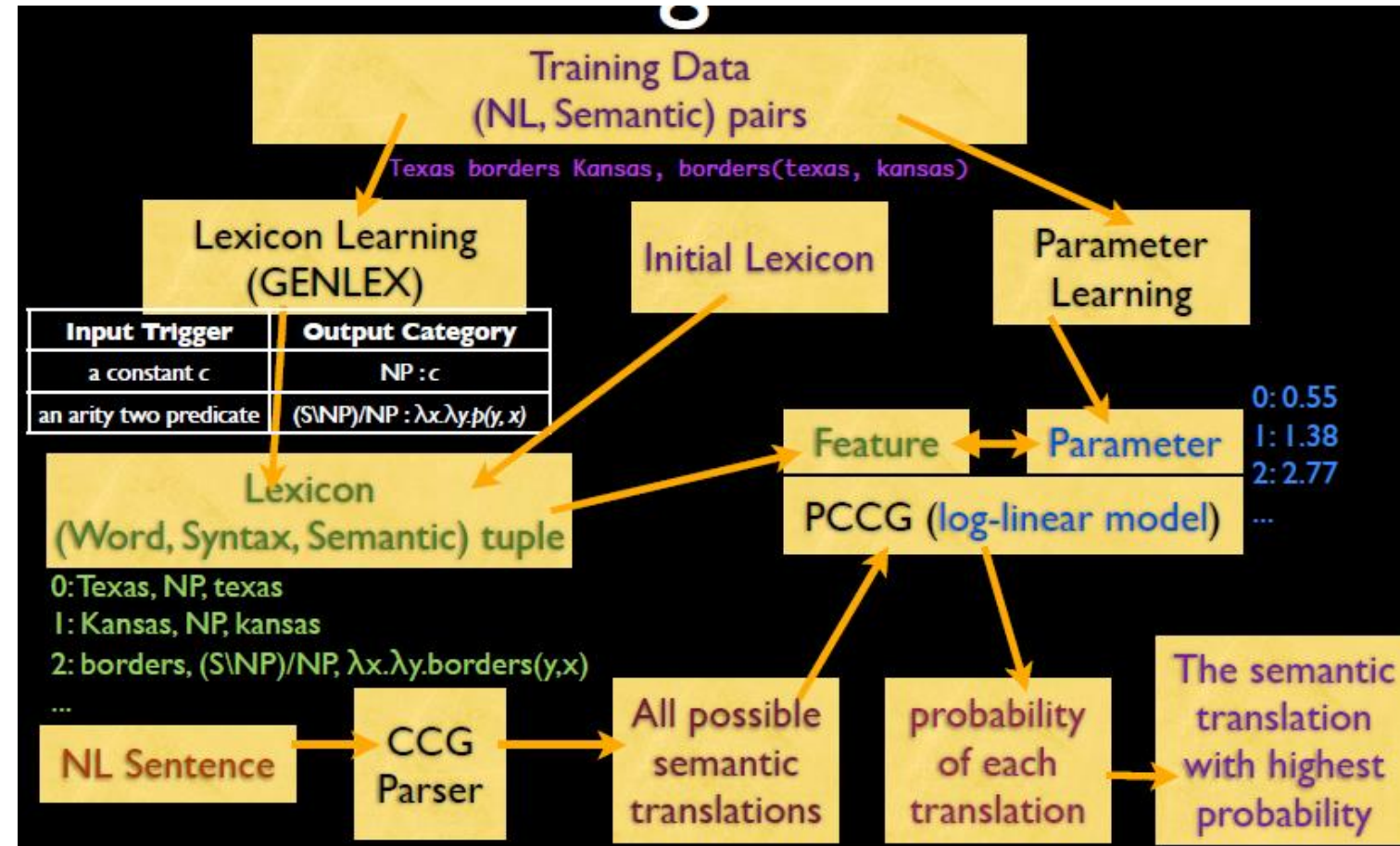


<http://m-stars.net/thinking-approaches-adaptive-to-square>

Learning to Map Sentences to Logical Form: Structured Classification with Probabilistic Categorical Grammars [Zettlemoyer and Collins 2005]

- Mapping from natural language to logical form as a structured prediction problem.
- Use lambda calculus to represent logical forms (semantics) and probabilistic combinatorial categorical grammar (PCCG) for syntax.

Good: More scalable and higher Accuracy compared to [Zelle and Mooney 1996]
Bad: Stick to a task/domain (supervised learning)



Learning to Map Sentences to Logical Form: Structured Classification with Probabilistic Categorical Grammars [Zettlemoyer and Collins 2005]

- Combinatorial Categorical grammar
 - Lexicons and parsing rules
 - Lambda functions and predicates

<u>Texas</u>	<u>borders</u>	<u>Kansas</u>
NP	(S\NP)/NP	NP
texas	<u>$\lambda x. \lambda y. \text{borders}(y, x)$</u>	<u>kansas</u>
S\NP		
<u>$\lambda y. \text{borders}(y, \text{kansas})$</u>		
S		
<u>$\text{borders}(\text{texas}, \text{kansas})$</u>		

Words	Category	
	Syntax	Semantic
Texas	NP	texas
border(s)	(S\NP)/NP	$\lambda x. \lambda y. \text{borders}(y, x)$
border	NP	$\lambda x. \text{border}(x)$
Kansas	NP	kansas
Kansas city	NP	kansas_city_MO

$X/Y : f \quad Y : a \Rightarrow X : f(a)$		
(S\NP)/NP	NP	S\NP
$\lambda x. \lambda y. \text{borders}(y, x)$	texas	$\lambda y. \text{borders}(y, \text{texas})$
$Y : a \quad X \backslash Y : f \Rightarrow X : f(a)$		
NP	S\NP	S
kansas	$\lambda y. \text{borders}(y, \text{texas})$	$\text{borders}(\text{kansas}, \text{texas})$

Learning to Map Sentences to Logical Form: Structured Classification with Probabilistic Categorical Grammars [Zettlemoyer and Collins 2005]

- Probabilistic Combinatorial Categorical grammar

- There can be many possible parses.
 - Lexicons L , steps T (hidden), a sentence S .
 - “John takes a plane.” – 3 possible parses
 - Some of the parses may be wrong.
 - Learn the best one using log-linear model optimized by Stochastic Gradient Descent (SGD).

$$\operatorname{argmax}_L P(L \mid S; \bar{\theta}) = \operatorname{argmax}_L \sum_T P(L, T \mid S; \bar{\theta})$$

+ [S]	Ey.(plane(y); take(#x.plane(x),y))
!+ [NP]	#x.plane(x)
!!+ [N]	#x.plane(x)
!+ [S\NP]	#z.Ey.(plane(y); take(z,y))
!!+ [(S\NP)/NP]	#w.#z.(w@ #x.take(z,x))
!!+ [NP]	#a.Ey.(plane(y); (a@ y))
!!!+ [NP/N]	#w.#a.Ey.((w@ y); (a@ y))
!!!+ [N]	#x.plane(x)

+ [S]	Ey.(plane(y); take(john,y))
!+ [NP]	john
!!+ [N]	john
!+ [S\NP]	#z.Ey.(plane(y); take(z,y))
!!+ [(S\NP)/NP]	#w.#z.(w@ #x.take(z,x))
!!+ [NP]	#a.Ey.(plane(y); (a@ y))
!!!+ [NP/N]	#w.#a.Ey.((w@ y); (a@ y))
!!!+ [N]	#x.plane(x)

+ [S]	Ey.(plane(y); take(#u.(u@ john),y))
!+ [NP]	#u.(u@ john)
!!+ [N]	#u.(u@ john)
!+ [S\NP]	#z.Ey.(plane(y); take(z,y))
!!+ [(S\NP)/NP]	#w.#z.(w@ #x.take(z,x))
!!+ [NP]	#a.Ey.(plane(y); (a@ y))
!!!+ [NP/N]	#w.#a.Ey.((w@ y); (a@ y))
!!!+ [N]	#x.plane(x)

Unsupervised Semantics Parsing

[Poon and Domingos EMNLP 2009 Best paper]

- Reduce supervision by learning target predicates and objects.
- Clustering of syntactic and semantic relations with the same meaning.
(distributional similarity)

BUYS (-, -)

= {*buys, acquires, 's purchase of, ...*}

= Cluster of various expressions for acquisition

Microsoft buys Powerset

Microsoft acquires semantic search engine Powerset

Powerset is acquired by Microsoft Corporation

The Redmond software giant buys Powerset

Microsoft's purchase of Powerset, ...

- Syntax then semantics not both at the same time.
 - Recursive clustering to produce parse trees then find the best corresponding semantic parse.

Good: Unsupervised

Bad: Require a lot of memory

```
java -mx20000m -cp usp.jar semantic.Parse genia results
```

<http://yoavartzi.com/sp14/slides/poon.sp14.pdf>

<http://alchemy.cs.washington.edu/usp/quick-start.pdf> <http://research.microsoft.com/en-us/um/people/hoifung/talks/usp09.pdf>

Unsupervised Semantics Parsing

[Poon and Domingos EMNLP 2009 Best paper]

- Pipeline
 - Dependency Tree → Quasi-logical Forms → lambda-forms → trees of lambda-forms → probabilistic distribution and MAP estimation over trees
- Markov Logic
 - (First order logic, real number weighting) tuples
- Evaluated on Question Answering in GENIA database
 - Comparable to the state-of-the-art TextRunner or DIRT system but answer a lot more of questions
 - Can learn passive voices, synonyms,...

Unsupervised Semantics Parsing

[Poon and Domingos EMNLP 2009 Best paper]

Q: What does IL-13 enhance?

A: The 12-lipoxygenase activity of murine macrophages

Sentence:

The data presented here indicate that (1) the 12-lipoxygenase activity of murine macrophages is upregulated in vitro and in vivo by IL-4 and/or IL-13, (2) this upregulation requires expression of the transcription factor STAT6, and (3) the constitutive expression of the enzyme appears to be STAT6 independent.

Combined Distributional and Logical Semantics [Lewis and Steedman 2013]

- Generalizability based on synonyms
 - $\text{Born_in}'(a, y) = \text{Birth_place}'(a, y) = \text{relation53}(a, y)$.

$\text{born} \vdash (\mathbf{S\backslash NP})/\mathbf{PP[in]} : \lambda x \lambda y . \text{relation53}(x, y)$

$\text{birthplace} \vdash \mathbf{N/PP[of]} : \lambda x \lambda y . \text{relation53}(x, y)$

- Learn to map multiple types to the same meaning.
 - Type clustering on entities by topic modeling (standard LDA).



Combined Distributional and Logical Semantics

[Lewis and Steedman 2013]

- Evaluation: generate 1000 question set from Gigaword Corpus and perform QA in the same way as [Poon and Domingos 2009].
- The follow up work [Lewis and Steedman 2014] includes intensional logic for machine reading so that it can handle time context.

Premises:

Every European has the right to live in Europe.
Every European is a person.
Every person who has the right to live in Europe can travel freely within Europe.

Hypothesis:

Every European can travel freely within Europe

Question: What did Delta merge with?

Sentence: The 747 freighters came with
Delta's acquisition of
Northwest

Answer: Northwest

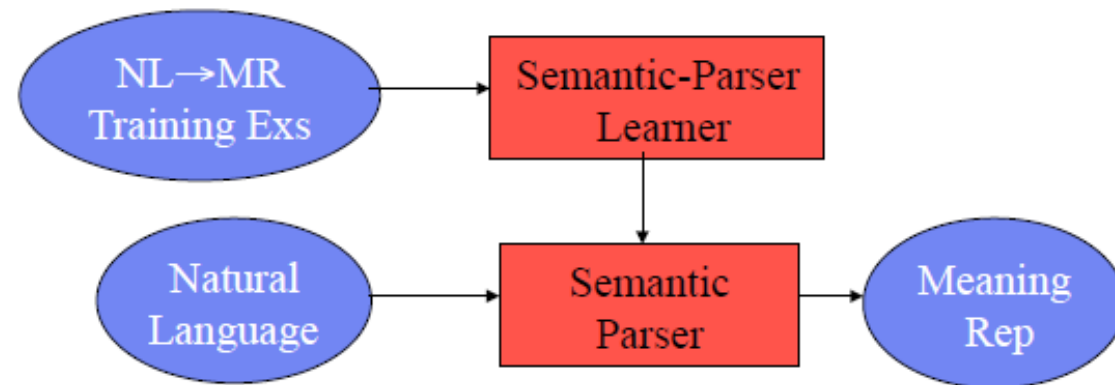
Good: Can model multiple premise sentences

Bad: A day to run on 12 cores
does not make use of existing ontologies

<http://homepages.inf.ed.ac.uk/s1049478/publications/tacl2013talk.pdf>

Conclusion

- A renaissance of an old theory when CCG is introduced in 2005 with good results [Zettlemoyer and Collins 2005].
- Can lead to a lot of interesting AI applications that provide rich semantics.



<http://yoavartzi.com/sp14/slides/mooney.sp14.pdf>

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- Dipanjan Das, Nathan Schneider, Desai Chen, and Noah A Smith. **Probabilistic frame-semantic parsing**. In Human language technologies: The 2010 annual conference of the North American chapter of the association for computational linguistics, pages 948-956. Association for Computational Linguistics, 2010. (replace logic with frame. Excluded. I thought there are some logics in frames.)

Requirements

- a) what is the problem being solved and why is it useful (i.e. why should anyone care)?
- b) how do the papers you studied address the problem?
- c) I'd like to see technical details with relevant math of at least one of the papers you studied
- d) what are the strengths and weaknesses of the work done in the suite of papers you studied?
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- 3) Your presentation should last for 25 minutes with 10 minutes left for questions. Here is a possible way to split up the time.
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- 3 min on: Item 2(a) above
- 7 min on: Item 2(b) above
- 10 min on: item 2(c) above
- 5 min on: item 2(d) above